

Quantum-Well Optimization for Blue InGaN/GaN Micro-LED Transmitters in Parallel Optical Systems

Jacob Torry

Department of Electrical and Computer Engineering

University of Illinois at Urbana-Champaign, Urbana, Illinois, 61801, USA

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Abstract

This project is motivated by the need for faster and more energy-efficient data transfer in modern computing and AI systems, where electrical interconnects are increasingly limited by bandwidth and energy per bit. Parallel arrays of blue InGaN/GaN micro-LEDs are a promising approach for short-reach optical communication, since many emitters can share the optical load while each pixel operates near its optimal current density. However, the performance of each pixel depends not only on the active-region design choices such as the number of quantum wells (QWs), but also on surface-related effects that introduce additional nonradiative recombination.

In this project, Advanced Physical Models of Semiconductor Device (APSYS) simulations are used to analyze blue InGaN/GaN micro-LEDs with varying QW counts. Radiative, Shockley-Read-Hall (including surface recombination), and Auger recombination mechanisms are evaluated, along with internal quantum efficiency as a function of current density. The results are then used to assess how device-level design choices impact system-level performance in parallel micro-LED arrays, highlighting trade-offs between efficiency, recombination behavior, and operating current density.

Introduction

Motivation

As computing systems continue to scale, data transfer is becoming a larger bottleneck. In AI systems especially, moving data between memory and compute can limit overall system performance. Over the past 20 years, peak server hardware floating-point operations per second (FLOPS) have scaled at 3.0x per two years, while DRAM and interconnect bandwidth have only scaled at 1.6x and 1.4x every two years, respectively [1]. This growing gap between compute and memory bandwidth is commonly referred to as the memory wall [1].

Electrical interconnects are still the standard, but they suffer from a number of issues as speed and density increase. At smaller dimensions, reducing wire width increases resistance, and increasing height to compensate raises capacitance between neighboring wires [2]. On top of that, electrical interconnects already consume roughly half of a microprocessor's power budget, and that number is projected to grow to around 80% [2]. These limitations make it harder to keep scaling performance with electrical links alone.

Because of this, optical interconnects are becoming more attractive. They avoid the RC delay and interference issues of metal wires, and have already demonstrated throughputs beyond 100Gbps [2]. Table 1 below summarizes some of the key differences between electrical and optical interconnects.

	Electrical	Optical
Bandwidth Scaling (per 2 years)	1.4x-16x	>100x demonstrated
Power Consumption	~80% of processor power	Target: 10-200 fJ/bit
Signal Integrity	RC delay, EM coupling	Immune to EM Interference

Table 1: Comparison of electrical and optical interconnects [1], [2].

Micro-LEDs are one possible optical transmitter technology. They can be made small, they can potentially switch fast, and they can be arranged in arrays for parallel communication. The idea is that instead of driving a single device hard, many pixels can share the load and each one operates closer to its efficiency peak.

Existing Work

There is already a significant amount of work on optical interconnects, particularly in silicon photonics, which has demonstrated complete interconnect systems ranging from 20 to 80 Gbps [2]. Vertical-cavity Surface-emitting lasers (VCSELs) are another common transmitter technology for short-reach links, though both silicon photonics and VCSELs require serialization/deserialization circuitry that takes up significant area and power [3]. More recently, micro-LEDs have been explored as an alternative. Parallel arrays of GaN micro-LEDs and micro-photodiodes have been demonstrated for short-range chip-to-chip communication, with the aggregated data rate scaling with the number of pixels rather than requiring a high-speed serial link per channel [3]. InGaN-based micro-LEDs have also been shown to work for visible-light communication, with yellow micro-LED arrays achieving data rates exceeding 1Gbps [4].

From the device side, InGaN/GaN LEDs have been studied heavily. A well-known issue is efficiency droop, where internal quantum efficiency peaks at moderate current densities and then falls off at higher injection levels, a behaviour linked to Auger recombination and carrier overflow [4]. In micro-LEDs specifically, surface recombination is an additional concern because the high perimeter-to-area ratio of small devices means sidewall damage can significantly increase nonradiative recombination. Bandwidth in these devices is also limited by carrier recombination lifetime, which is affected by the internal electric field in GaN quantum wells [3]. Prior work has looked at how design parameters like QW count and indium content affect these behaviors, but there is less focus on how these choices translate to system-level performance in parallel low-current arrays.

Device Physics and Background

Light Emitting Diode Structure

An InGaN/GaN micro-LED is essentially a p-n junction with one or more InGaN quantum wells in the active region, sandwiched between n-type and p-type GaN layers. When a forward voltage is applied, electrons from the n-side and holes from the p-side get injected into the active region and recombine. Some of this recombination is radiative and produces light, while the rest is nonradiative and just wastes energy as heat, which tends to hurt the overall efficiency of the device.

The reason modern LEDs use this sandwich heterostructure design rather than a simple homojunction comes down to carrier confinement. In a homojunction there is nothing stopping carriers from diffusing away from the junction before they recombine, which leads to poor efficiency. By instead using two materials with different bandgaps, a heterojunction creates a potential well that traps carriers in a narrow active region and forces recombination where carrier density is highest [5]. In the InGaN/GaN system, the narrow-bandgap InGaN wells provide exactly this confinement, while the wider-bandgap GaN layers on either side act as barriers. This is illustrated in figure 1 below.

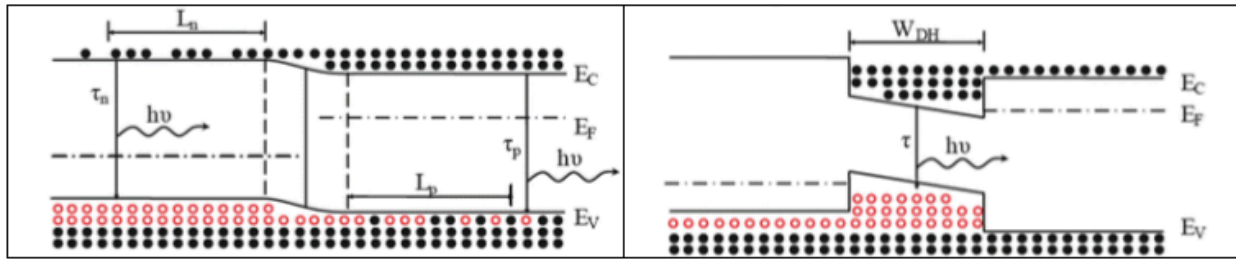


Figure 1: Homojunction (Left) and Heterojunction (Right) under forward bias [6].

Modern devices extend this further with multiple quantum wells (MQWs), increasing the total volume available for radiative recombination while maintaining confinement. However, more wells also complicate carrier transport, making uniform carrier distribution across the MQW stack an important design consideration [5].

Operation Physics

The emission wavelength of an InGaN/GaN LED depends on the bandgap of the active region material. For an alloy like $In_xGa_{1-x}N$, the band gap can be estimated using a modified version of Vegard's law that includes a bowing correction as shown in equation 1.

$$E_g^{In_xGa_{1-x}N}(x) = x * E_g^{InN} + (1 - x) * E_g^{GaN} - b * x(1 - x) \quad (1)$$

Where the bandgaps of GaN and InN at room temperature are about 3.4 eV and 0.7 eV respectively, and b is a bowing parameter that is determined from experimental data [5]. Once the bandgap is known, the corresponding emission wavelength is given by equation 2.

$$\lambda_0 = \frac{1240}{E_g(eV)} (nm) \quad (2)$$

So by tuning the indium content x , you can shift the emission wavelength across a wide range. For blue emission around 450nm, you need a bandgap of about 2.76 eV, which works out to an indium fraction somewhere in the 15 – 20% range depending on the bowing parameter used.

Internal Quantum Efficiency

The IQE comes down to how well the device can promote radiative recombination over the competing nonradiative forms. The three main recombination mechanisms considered in this report are radiative recombination, Shockley-Read-Hall (SRH) recombination, and Auger recombination. Using the standard ABC model, IQE can be expressed as such in equation 3.

$$IQE = \frac{Bn^2}{An+Bn^2+Cn^3+f(n)} \quad (3)$$

Where n is the carrier concentration, A is the SRH coefficient, B is the radiative coefficient, C is the Auger coefficient, and $f(n)$ accounts for carrier leakage out of the active region [5]. It is imperative to realize all three mechanisms compete with each other, and improving IQE means pushing down the nonradiative contributions while keeping radiative recombination high.

This is where quantum wells become really important. By confining carriers spatially, quantum wells increase the local carrier density and make radiative recombination more likely. Adding more quantum wells isn't always better, since it can also change how evenly carriers are distributed across the active region, so the optimal number of wells is something that has to be figured out for each specific design.

Device Design

Structure and Doping

This project simulates blue InGaN/GaN micro-LED structures using APSYS, a commercial semiconductor device simulation software. The simulated device consists of an n-GaN base layer, followed by an InGaN/GaN multiple quantum well active region, and capped with a p-GaN layer. Two p-GaN layers are used on top - a lightly doped layer and a heavily doped contact layer. The device geometry is split into two columns: the active mesa (column 1)

and a passivated sidewall region (column 2). The active region consists of InGaN quantum wells with an indium composition of $x=0.194$, targeting blue emission near 450nm , separated by GaN barrier layers.

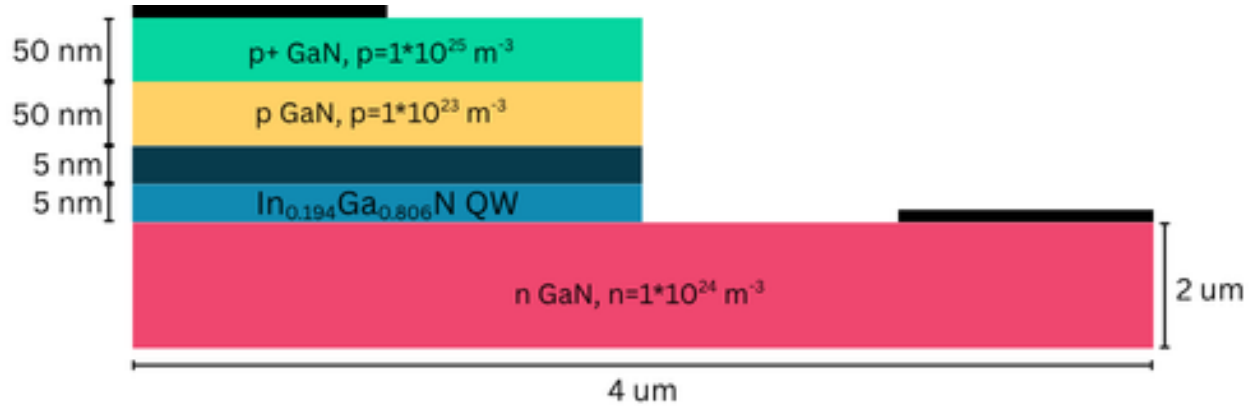


Figure 2: Micro LED structure used in simulation (with varying QW counts)

Simulating a 2QW structure resulted in the band structure shown in figure 3 below which clearly shows the valence band (blue) and conduction band (red), but more importantly shows 2 distinguishable quantum wells that have a band gap roughly equal to 2.6 eV which was the target value to produce 405nm blue light.

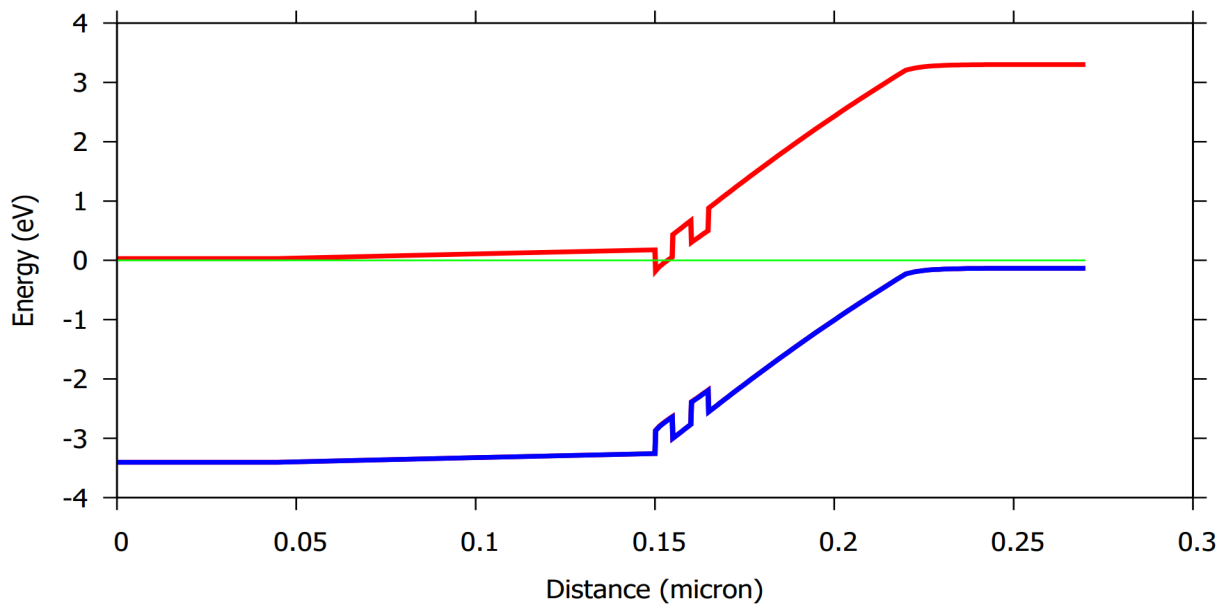


Figure 3: Band Structure of a 2 QW micro LED

Recombination Assumptions

The recombination parameters used in the simulations are set specifically in the .sol file. The Auger coefficients for electron and holes in the InGaN active region are both set to $3 * 10^{-42} m^3/s$, and the radiative recombination coefficient is set to $6.651 * 10^{-16} m^3/s$. These values were not changed during the simulations, however, we did vary the surface recombination parameters. Surface recombination is one of the defining challenges of micro-LEDs compared to larger devices, since the surface-to-volume ratio increases significantly as device size shrinks. In this simulation, surface effects are modeled along the mesa sidewall using two interface conditions. First, an acceptor-type trap is defined with a trap density of $1 * 10^{16} cm^{-3}$, a trap energy level of $1.4eV$, and a carrier lifetime of $1 * 10^{-6} s$ for both electrons and holes. In order to study how the effects of sidewall recombination affects efficiency; three cases were tested by varying surface recombination velocity of carriers as described in table 2.

	Weak Recomb. Velocity	Strong Recomb. Velocity
Velocity_n [cm/s]	$1 * 10^2$	$1 * 10^5$
Velocity_p [cm/s]	$1 * 10^2$	$1 * 10^5$

Table 2: Varying Surface Recombination Velocities for Surface Effects

Sweeping conditions

The simulation runs in two stages. First, a voltage sweep is performed from $0V$ up to $10V$ to bring the device to target current density of approximately $0.01 A/m$, using an initial step of $0.001V$ and a maximum step of $0.01V$. From the pre-converged solution, a current sweep then drives the device from that point up to $-50A/m$ with a maximum step of $0.2 A/m$, capturing the full current-dependent behavior of internal quantum efficiency.

Parameter Table

n-GaN Doping	$1 * 10^{24} m^{-3}$
p-GaN Doping	$1 * 10^{23} m^{-3}$
P+ GaN Contact Doping	$1 * 10^{25} m^{-3}$
Indium Composition	0.194
Target Emission Wavelength	450 nm
QW Thickness	5 nm
Barrier Thickness	5 nm
QW Count (swept)	1, 2, 3, 4
Radiative Coefficient	$6.651 * 10^{-16} m^3/s$
Auger Coefficient (n and p)	$3 * 10^{-42} m^6/s$
Trap Density	$1 * 10^{16} cm^{-3}$
Trap Energy Level	1.4 eV
Carrier Lifetime (Trap)	$1 * 10^{-6} s$
Surface Recomb. Velocity (weak)	$1 * 10^2 cm/s$
Surface Recomb. Velocity (strong)	$1 * 10^5 cm/s$
Voltage Sweep Range	0 → 10 V
Current Sweep Range	0 → 50 A/m

Table 3: Summary of Key Simulation Parameters

Simulation Results

Recombination Analysis

Figure 4 shows the fractional contributions of radiative, SRH, and Auger recombination as a function of current density for a 4 quantum well LED, illustrating the shift in the dominant mechanism with increasing injection.

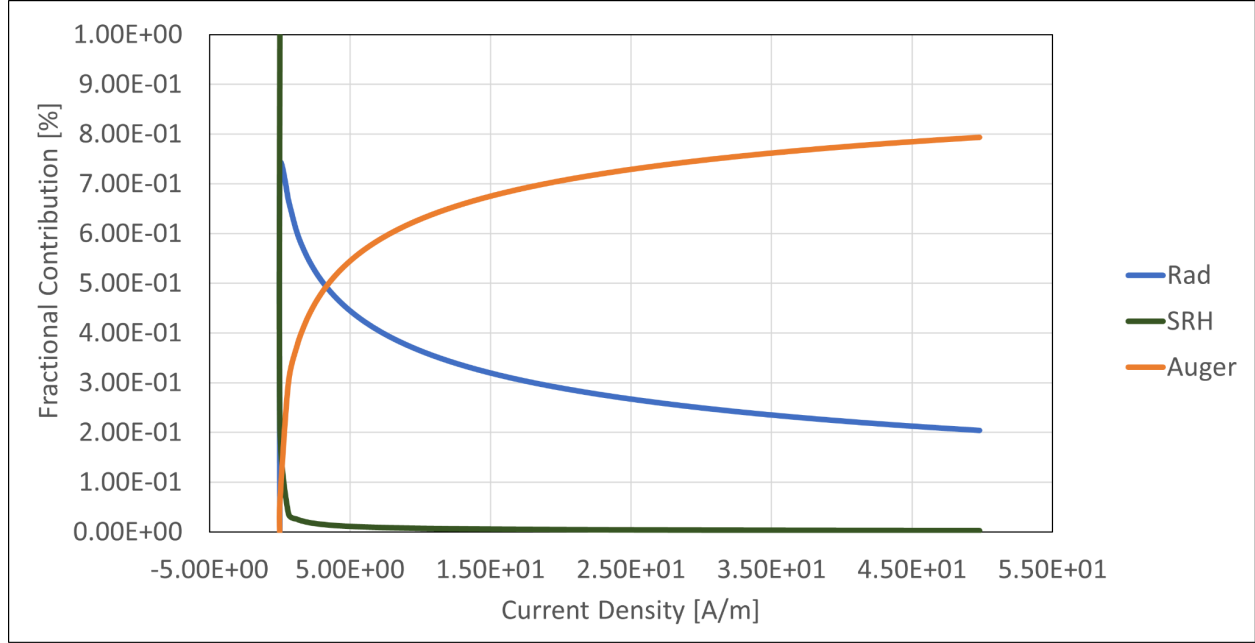


Figure 4: Fractional Contribution from each Recombination Mechanism in a 4QW LED

At the onset of current injection, SRH recombination accounts for nearly 100% of all recombination, but collapses sharply within an extremely narrow current density range. This is expected - at low carrier densities the linear An term from equation 3 dominates, but as injection increases radiative and Auger recombination grow faster and SRH becomes negligible. After SRH collapses, radiative recombination briefly becomes the dominant mechanism, which corresponds to the peak efficiency window in the IQE curves. However, because Auger recombination scales as Cn^3 while radiative only scales at Bn^2 Auger grows faster and passes radiative. By 50 A/m Auger accounts for roughly 80% of all recombination, which is the primary driver of efficiency droop at high current densities.

Efficiency

Figures 5, 6, and 7 show IQE as a function of current density for the no-trap, moderate-trap, and strong trap surface recombination cases.

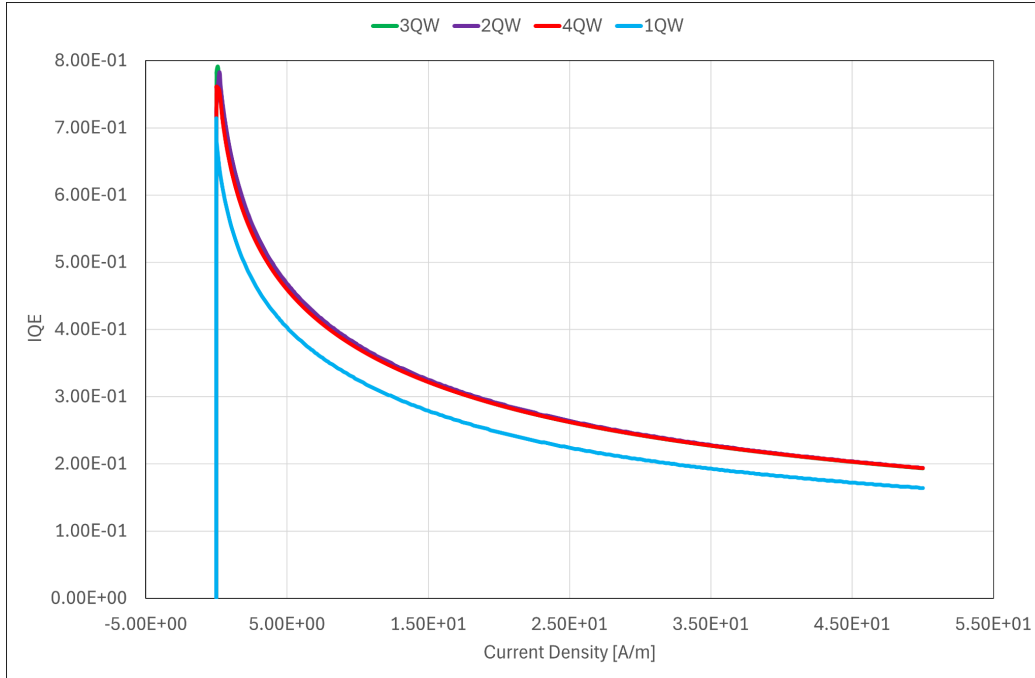


Figure 5: IQE vs Current Density for 1-4 QW devices with No Surface Effects

After simulating and collecting the IQE with no surface effects, interfaces were added and we changed surface recombination to moderate velocities according to table 2.

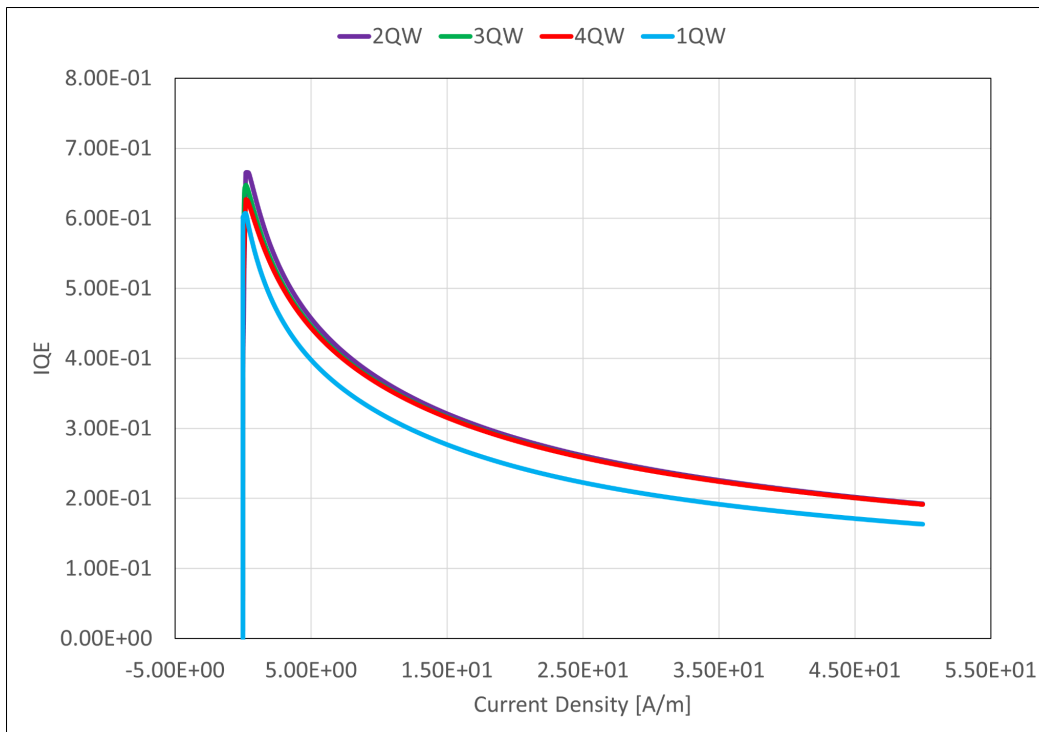


Figure 6: IQE vs Current Density for 1-4 QW devices with Moderate Surface Effects

Again, once IQE data was collected, surface recombination velocity was increased to simulate strong/severe surface losses.

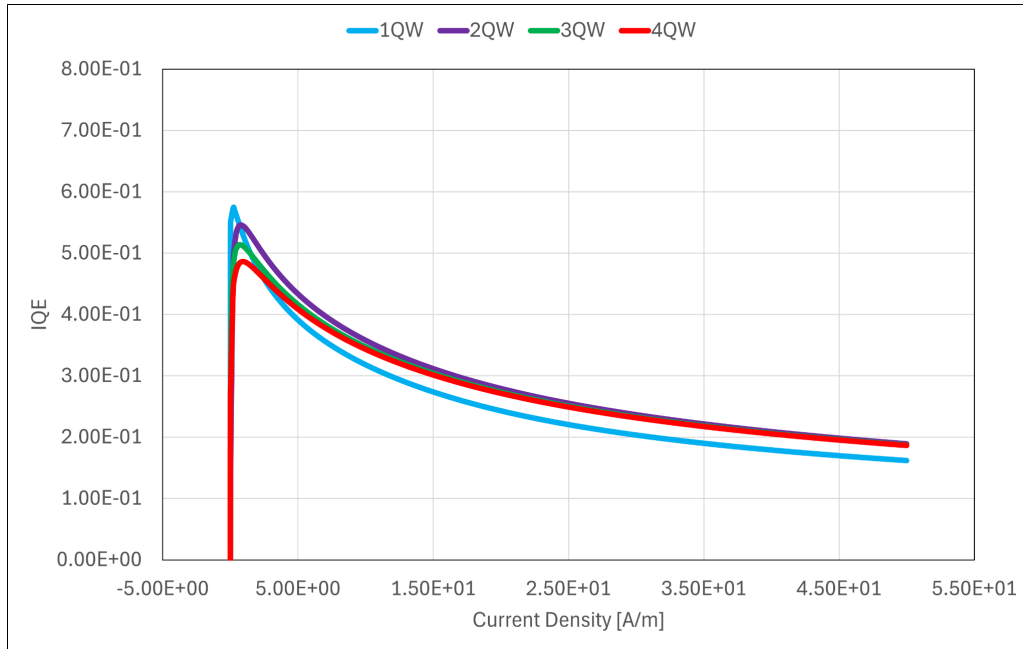


Figure 7: IQE vs Current Density for 1-4 QW devices with Strong Surface Effects

The effect of surface traps is most pronounced at low current densities, where IQE is already being suppressed by SRH recombination. Surface traps introduce additional nonradiative recombination pathways at the mesa sidewalls, which effectively add to the A coefficient in equation 3 and pulls the IQE curve down across the board. The peak IQE drops and shifts slightly toward higher current density as trap density increases, since higher injection is needed before radiative recombination can overcome the combined SRH and surface losses. This is directly relevant to micro-LED scaling. As device diameter shrinks, the sidewall surface area becomes a larger fraction of the total active region, so surface recombination becomes increasingly damaging to efficiency.

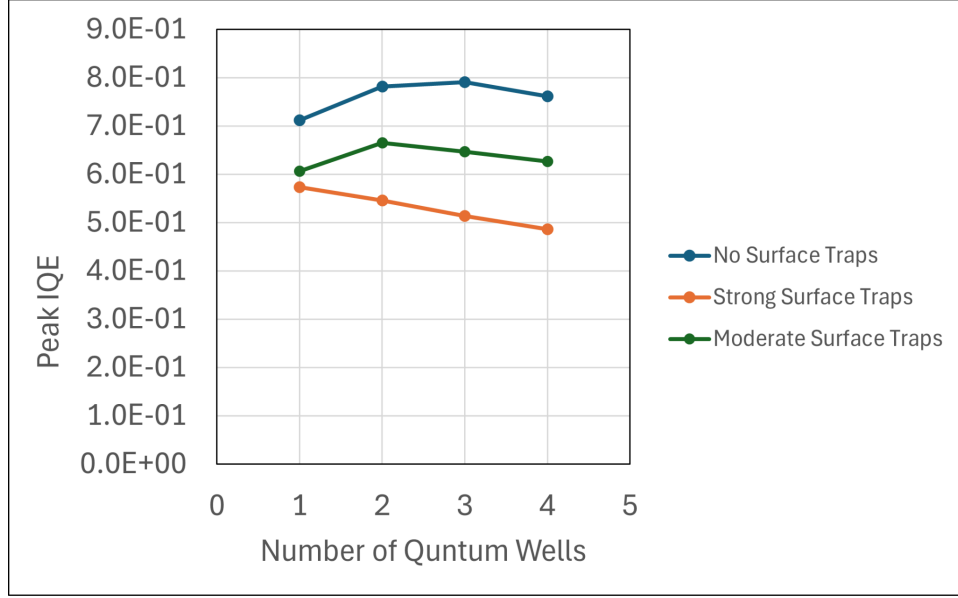


Figure 8: Peak IQE as a Function of Quantum Well Count

Figure 8 shows peak IQE as a function of QW count for all three surface trap conditions. Across all three cases, peak IQE is not just increasing with well count. Instead it peaks at 2-3 quantum wells and then drops off slightly at 4 wells. This suggests that while adding wells initially helps by increasing the active volume available for radiative recombination, beyond a certain point the benefits plateau. This is potentially due to carrier transport limitations. If carriers injected are not making it all the way to all the wells, then we are under utilizing the wells which would drag down efficiency.

The gap between the three curves is also telling. Without surface traps, peak IQE reaches about 79% at 3 wells, while the moderate trap case peaks around 66% at 2 wells, and the strong trap case drops to around 57% at 1 well and falls consistently from there. This suggests that when surface recombination is strong, adding more wells does not help and may actually hurt efficiency, possibly because the increased device volume means more sidewall exposure relative to the radiative recombination gained.

Parallel Array Analysis

The broader motivation for this project is not only improving a single pixel, but also understanding what kind of pixel design might be useful in a parallel micro-LED transmitter array. A simple idea is that instead of driving one device really hard, a system could use many smaller devices at lower current.

$$P_{tot} = N * P_{pixel}$$

If each device operates close to its peak efficiency, the overall system may perform better than a single device driven into the Auger-dominated high-injection regime.

The simulation results directly inform this trade-off. From figure 4, the radiative-dominant window occurs at relatively low current densities before Auger recombination takes over. This means that for an array application, it is actually advantageous to keep each pixel operating at moderate injection rather than pushing for maximum optical output from a single device. The IQE curves in figures 5-7 reinforce this point, showing that peak IQE is reached at low to moderate current densities across all surface trap conditions, after which droop sets in regardless of well count or surface quality.

Figure 8 adds another layer to this; the 2-3 QW designs consistently achieve the highest peak IQE across all trap conditions, meaning they represent the best candidate pixel design for an array where each device is intended to operate near its efficiency peak. A single heavily driven device with more wells does not compensate for the efficiency lost to Auger droop, whereas an array of optimized 2-3 QW pixels operating in the radiative-dominant regime could collectively deliver more light at higher overall efficiency.

Conclusion

This project investigated how quantum well count and surface recombination affect the efficiency of blue InGaN/GaN micro-LEDs for potential use in parallel optical interconnect systems. Using APSYS simulations, devices with 1-4 quantum wells were compared across three surface trap conditions in terms of recombination behavior, peak IQE, and efficiency droop.

The recombination analysis showed that SRH dominates at low current densities but collapses quickly as injection increases, after which radiative recombination briefly dominates before Auger takes over at high current densities. This Auger-driven droop sets in relatively early, meaning devices spend most of their operating range in an efficiency-degraded condition if driven hard.

In terms of quantum well count, peak IQE was not consistently increasing. Instead it peaked at 2-3 wells before declining slightly at 4 wells, suggesting that carrier transport limitations and surface recombination begin to outweigh the benefits of additional active volume beyond a certain point. Surface recombination consistently reduced peak IQE across all well counts, and was severe enough in the strong-trap case to eliminate the efficiency benefit of adding more wells entirely.

Taken together, these results suggest that a 2-3 QW design operating at moderate current density represents the most efficient pixel configuration for a parallel array application. Rather than driving a single device hard into the Auger-dominated regime, distributing optical output across multiple optimized pixels operating near their efficiency peak is a more promising approach for high-efficiency optical interconnect systems. Surface quality remains a vital limitation, and reducing sidewall recombination will be critical to realizing the full efficiency potential of micro-LED array designs at small device sizes.

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